3DGENGINE / EDITOR WORKFLOW GUIDE

Animation Preview Panel Guide

Build locomotion, actions, events, state graphs, layered animation, and root-motion behavior.

INSIDE THIS GUIDE

Fast Idle / Walk / Run setup
State Graph parameters and transitions
Full-body and masked Action Profiles
Animation Events for attacks, audio, and effects
Runtime diagnostics and script integration

Current engine workflow reference | July 2026

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Quick start

Use this path when the model already contains Idle, Walk, and Run clips and you want a working controller quickly.

1 **Select the animated object**
Choose the skeletal model object in Hierarchy. In Inspector > Animation, enable Skeletal Model and choose a Default Clip.

2 **Assign basic locomotion**
Open Inspector > Animation > Locomotion States. Enable Use Locomotion; choose Idle, Walk, and Run; set Walk At and Run At.

3 **Open the preview**
Choose Panels > Animation Preview. Use Preview Clip, Play/Pause, Time, and Reset to inspect every clip.

4 **Test movement**
Enter Play mode. The built-in player controller writes Speed from movement input and sprint speed.

5 **Save the scene**
Save after authoring. The scene stores clip indices and names, graph data, events, and action profiles.

KEY RULE If State Graph contains one or more states, it is used instead of the simple Use Locomotion setup. Keep the graph empty for the quick path.

Recommended starter values

Setting	Value	Reason
Walk At	0.15	Avoid tiny input noise.
Run At	3.0	Matches the default threshold.
Playback Speed	1.0	Preserves authored timing.
Transition Fade	0.15-0.25 s	Smooth and responsive.

Know the panel

Author and preview in Edit mode; inspect the live controller in Play mode. The selected Hierarchy object drives the panel.

Section	Purpose
Preview Controls	Play/Pause, Reset, Preview Clip, scrub Time, and read duration.
Authored	Read-only summary of the default clip and simple locomotion.
Action Test	Try a clip as full body or from a Mask Root Bone.
State Graph	Create states, parameters, transitions, blend clips, and root motion.
Graph View / Validation	See connectivity and resolve authoring warnings.
Graph Parameters	Change preview Float, Bool, and Trigger values in Edit mode.
Transition Debug	See why a Play-mode transition is blocked, ready, or selected.
Action Profiles	Save reusable actions that scripts play by name.
Events	Place named markers at clip times; audio commands are supported.
Runtime / Clips / Skeleton	Inspect live playback, clip inventory, and the bone hierarchy.

SELECTION DRIVES THE PANEL

If the panel says no animated model is selected, select the skeletal object itself, not a weapon child, collider, or ground object.

Prepare the character

- Enable Skeletal Model and confirm the Inspector reports non-zero Clips and Bones.
- Preview every clip at speed 1.0; check orientation, scale, deformation, loop seam, duration, and root drift.
- Use clear clip names: Idle, Walk, Run, Attack, Cast, Hit, and Death.
- Open Skeleton and locate the spine/chest bone for upper-body masks.
- Fix a model that faces down or moves on the wrong basis before tuning animation.

Preview clips precisely

Preview Controls lets you inspect the source clips before building any runtime rules.

1. **Choose Preview Clip**
   Selecting a clip resets preview time. Empty clip names appear as Clip N.
2. **Play or pause**
   Pause before placing an event or checking a contact pose.
3. **Scrub Time**
   Drag from 0 to clip duration. Reset returns to 0 seconds.
4. **Tune playback carefully**
   Inspector playback speed, state speed, and action speed are separate multipliers.
5. **Inspect bones**
   Hover a bone to see its parent. Selecting one can fill Action Test Mask Root Bone.

What to inspect

Type	Look for	Loop
Idle	Breathing extremes, stable hands/staff	Yes
Walk / Run	Heel strikes, stride length, root drift	Yes
Attack / Cast	Wind-up, release/impact, recovery	No
Jump	Takeoff, apex, landing	Usually split
Hit / Death	Readability, contact, final pose	No

TIMING DISCIPLINE Add Event uses the current Preview Clip index and current preview time. Pause and scrub to the exact pose first.

Two ways to build Idle - Walk - Run

The quick helper is ideal for three clips. The State Graph is better when locomotion interacts with jump, falling, attacks, hit reactions, or death.

Speed range	Result	Starter rule
Speed < Walk At	Idle	below 0.15
Walk At <= Speed < Run At	Walk	0.15 to 3.0
Speed >= Run At	Run	3.0 or more

Quick helper workflow

- Inspector > Animation: enable Skeletal Model, Autoplay, and Loop; set Default Clip to Idle.
- Locomotion States: enable Use Locomotion and assign Idle, Walk, Run.
- Set Walk At = 0.15 and Run At = 3.0, then tune against real movement speed.
- In Play mode, move with WASD and sprint with Shift; Speed should return to 0 when input stops.

DO NOT DUPLICATE CONTROLLERS Authored graph states take priority. Remove graph states to return to the quick helper.

State Graph workflow

1. **Add parameters**
   Speed (Float), Grounded (Bool), and optional Jump/Attack (Trigger).
2. **Add states**
   Add Idle first, then Walk and Run. The first state becomes the initial state.
3. **Add both directions**
   Build Idle <-> Walk and Walk <-> Run. Every outward route needs a return.
4. **Set fades**
   Use about 0.15-0.25 s for locomotion.
5. **Validate**
   Graph Validation should say Graph is valid; Graph View should show all links.

Parameters, transitions, and blend clips

Type	Best use	Script call
Float	Speed, aim weight	SetAnimationParameter("Speed", value)
Bool	Grounded, Dead	SetAnimationBool("Grounded", value)
Trigger	Jump, Attack, Hit	SetAnimationTrigger("Attack")

Transition fields

- From can be a named state or Any State; To must be a real destination.
- Float comparisons use >= or <. Bool uses == or != with Expected. Trigger is consumed after firing.
- Fade controls crossfade duration. Exit Time is normalized 0-1; 0 means no wait.
- Higher Priority wins when several rules pass. Interrupt Blend allows a rule during an existing blend.

From	To	Condition	Exit	Priority	Interrupt
Any State	Attack	Attack trigger	0.00	20	Yes
Attack	Idle	Speed < 0.15	0.90	10	No
Attack	Walk	Speed >= 0.15	0.90	10	No

Blend Clip inside a state

Choose a main Clip, optional Blend Clip, a Float Blend Parameter, and Blend Min/Max. Example: Walk blended to Run with Speed from 1.5 to 5.0. Values between the bounds create a continuous blend weight.

GRAPH STATE OR ACTION PROFILE?

Use a graph state for long-lived logic. Use an Action Profile for a one-shot montage-like action that returns automatically to the controller.

Montage-like actions and layering

Action Profiles are reusable one-shot animations. Scripts play them by name while the state controller continues underneath.

Field	Meaning	Starter
Name	Stable script-facing identifier	StaffAttack
Clip	Attack, cast, reload, hit, etc.	Attack_Staff
Mask Root	Empty is full body; a bone masks to its subtree	Spine/chest
Fade In	Blend into action	0.08 s
Fade Out	Return to controller	0.15 s
Speed	Action playback multiplier	1.0

Full body vs layered

Mode	Mask Root	Movement	Best for
Full body	Full Body / empty	Locked until action completes	Heavy attack, death, committed cast
Layered	Spine/chest bone	Locomotion continues	Staff fire, reload, aim, light cast

- Use Action Test to choose clip, fades, speed, and mask; click Play Action and watch Status.
- When it feels correct, click Add Profile and give it a unique name.
- Use Pick Bone instead of typing. A wrong bone name can make the mask ineffective.
- Call PlayAnimationProfile("StaffAttack") from a script.

MONTAGE BEHAVIOR

A full-body profile has an empty mask. It pauses built-in player and AI movement until the action completes, then reveals the underlying state controller again.

Time gameplay and choose who moves the character

Animation Events

1. **Preview and pause**
   Choose the attack/cast clip and scrub to impact, contact, or release.
2. **Add Event**
   The panel captures the current clip index and time. Rename it immediately.
3. **Use stable names**
   Examples: StaffRelease, MeleeImpact, Footstep_L, Footstep_R.
4. **Consume once**
   WasAnimationEvent(name) is true when the marker is delivered in the current update.

EVENT-DRIVEN GAMEPLAY

```
void OnUpdate(float dt) override {
    if (WasAnimationEvent("StaffRelease")) {
        BurstParticles(24);
        PlayAudio(true);
        // Spawn the projectile/template here.
    }
}
```

AUDIO COMMAND NAMES

Events also support Audio.Play and targeted commands such as Audio.Restart:SourceName, Audio.Pause:SourceName, Audio.Resume:SourceName, and Audio.Stop:SourceName.

Root motion and movement ownership

Strategy	Root Motion	Movement owner
In-place locomotion	Off	Player controller, AI steering, or script
Authored displacement	On per state	Animation root delta
Full-body action	Profile mask empty	Movement locked until action ends
Layered action	Profile mask is a bone	Controller continues movement

Debug before you tune

Section	Watch	Meaning
Runtime	State, clips, times, blend, pose bones	Confirms active state and crossfade.
Graph View	Green active state	Shows the runtime destination.
Graph Parameters	Live Speed/bool/trigger	Confirms gameplay writes the parameter.
Transition Debug	Condition, Exit, Blend, Ready, Selected	Explains exactly why a rule did or did not fire.
Graph Validation	Edit-mode warnings	Finds broken states, parameters, and destinations.

Transition Debug status

Status	Meaning
Condition	The parameter comparison is false.
Exit	Condition passed; normalized exit time has not been reached.
Blend	An active blend blocks the rule unless Interrupt Blend is enabled.
Ready	Eligible, but may lose to a higher-priority transition.
Selected	This rule won and will run.

FAST DEBUGGING LOOP

Select the runtime character, watch Speed while moving, find the blocked transition row, then change only the field named by its status.

Common symptoms

Symptom	Fix
Locomotion settings ignored	State Graph has states; use it or remove every graph state.
Always idle	Watch Speed and compare it with Walk At.
Action stops movement	Choose a spine/chest Mask Root for a layered action.
Transition waits	Lower Exit Time or enable Interrupt Blend after reading debug.
Foot sliding	Match gameplay speed to stride, or use root motion intentionally.
Character travels twice	Disable Root Motion when controller/navmesh already moves it.

Drive animation from gameplay code

COMPLETE PATTERN

```
class StaffCombat : public engine::Script {
public:
    void OnUpdate(float dt) override {
        SetAnimationParameter("Speed",
            GetFieldFloat("movement_speed", 0.0f));
        SetAnimationBool("Grounded",
            GetFieldBool("grounded", true));

        if (WasMouseButtonPressed(0))
            PlayAnimationProfile("StaffAttack");
        if (WasKeyPressed('J'))
            SetAnimationTrigger("Jump");

        if (WasAnimationEvent("StaffRelease")) {
            BurstParticles(30);
            PlayAudio(true);
        }
        if (IsAnimationMovementLocked())
            return; // skip custom translation
    }
};
```

API	Purpose
SetAnimationParameter(name, float)	Write Float graph parameters.
SetAnimationBool(name, bool)	Write Bool graph parameters.
SetAnimationTrigger(name)	Fire a one-shot Trigger transition.
PlayAnimationProfile(name)	Play an authored full-body or masked action.
WasAnimationEvent(name)	React when a named marker fires.
IsAnimationMovementLocked()	Detect an active full-body action.

INPUT RULE Use WasKeyPressed or WasMouseButtonPressed for one-shot actions. Calling PlayAnimationProfile every frame continually restarts the animation.

Magic staff character: locomotion plus combat

1. **Locomotion**
   Create Idle, Walk, Run and Float Speed. Add two-way transitions at 0.15 and 3.0.
2. **Jump**
   Add Jump as non-looping, Trigger Jump, and Any State -> Jump.
3. **Staff attack**
   Create StaffAttack. Empty mask gives a committed full-body attack; a spine mask allows movement.
4. **Release event**
   Scrub to the frame where magic leaves the staff and add StaffRelease.
5. **Effects**
   On StaffRelease, burst particles, play audio, and spawn the projectile/template.
6. **Test**
   Use Action Test, then Play-mode Runtime and Transition Debug; verify return to locomotion.

Suggested graph data

Item	Configuration
Speed	Float, default 0
Grounded	Bool, default true
Jump	Trigger
Idle -> Walk	Speed >= 0.15, fade 0.20
Walk -> Idle	Speed < 0.15, fade 0.20
Walk -> Run	Speed >= 3.0, fade 0.18
Run -> Walk	Speed < 3.0, fade 0.18
Any -> Jump	Jump trigger, priority 20, interrupt on

DESIGN CHOICE Use an Action Profile rather than an Attack graph state when you want montage-like movement locking/layering and automatic return to whichever locomotion state was active.

11 / FINAL PASS

Ship-ready checklist

1. Skeletal Model is enabled and the object reports non-zero clips and bones.
2. Idle, Walk, and Run loop cleanly with consistent orientation.
3. Only one locomotion path is active: quick helper or State Graph.
4. State and parameter names are unique, stable, and correctly cased.
5. Every locomotion route has a return transition.
6. Graph Validation reports Graph is valid.
7. Full-body profiles intentionally lock movement; masked profiles preserve it.
8. Events are on the correct clip/time and tested at action speed.
9. Root Motion is enabled only where animation should own translation.
10. Runtime, Graph Parameters, and Transition Debug show expected values.
11. Scripts use one-shot input and exact profile/event names.
12. The scene is saved and the runtime export reproduces editor behavior.

RECOMMENDED WORKFLOW Preview clips -> author controller -> validate graph -> test actions/events -> inspect runtime -> save scene.

Panel path

Panels > Animation Preview | **Inspector > Animation** | Hierarchy selection drives both panels.